

Separating Mixtures — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, name the **property that differs** between the substances — soluble or insoluble, boiling point, magnetic, particle size — because that property is why the method works.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe mixtures or experiments you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- Naming a method scores one mark. Saying why it works is where the rest are.

SECTION 1 – UNFAMILIAR MIXTURES

- 1.** For each mixture below, choose a method, and explain the property it uses. Each part is worth three marks: one for the method, two for the reason.

(a) Copper sulfate crystals dissolved in water — you want the crystals back. [3]

(b) Chalk powder (insoluble) stirred into water. [3]

(c) Ethanol (boiling point 78 °C) mixed with water (boiling point 100 °C) — you want the ethanol. [3]

SECTION 2 – CHROMATOGRAPHY OF AN UNKNOWN

- 2.** A green food colouring is tested. On the chromatogram the solvent front travels **7.5 cm**. The colouring gives two spots: a blue one 4.5 cm from the baseline and a yellow one 1.5 cm from the baseline. A known dye, Tartrazine, run alongside, gives a yellow spot at 1.5 cm.

(a) Calculate the R_f values of the blue and yellow spots. [2]

(b) State, with a reason, whether the colouring contains Tartrazine. [2]

(c) The blue dye travelled much further than the yellow. Explain why, in terms of the dyes and the solvent. [2]

- 3.** Explain why R_f values are always less than 1, and why a student who calculates an R_f of 1.5 must have made a mistake. [3]

SECTION 3 – WHEN IT GOES WRONG

- 4.** A student sets up chromatography with the baseline drawn in ink and the solvent level above the baseline. Explain what goes wrong in each case. [4]

- 5.** A student distils ink to obtain water, but puts the thermometer bulb in the liquid instead of at the top of the flask. State what the thermometer reads compared with 100 °C, and explain why this matters. [3]

- 6.** A student tries to separate salt from sea water by filtering and is surprised the filtrate still tastes salty. Explain what has happened and what she should do instead. [3]

- 7.** Explain why distillation needs far more energy than filtration, even though both can separate a solid from a liquid. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.